

System Evaluation and Next Steps

Part 1: Work Done

System Enhancement

The system was enhanced by adding two components:

- Symbolic verification
- Logical inconsistency detection

We compared five models (GLM-5, LLAMA3.1:70b, Gemini2.5-flash, GPT4.1, Mistral-large-latest) overall and determined the best model for each part by explicit testing. The best model was fixed for each part, then the next pipeline stage was tested.

Answer & Extraction

Tested on 5 questions covering the whole context. Results for triple extraction:

Table 1: Extraction results per model (Extracted / Gold-correct / Extra-correct / Wrong)

Model	Q1	Q2	Q3	Q4	Q5
GPT-4.1	3/3/0/0	3/3/0/0	3/3/0/0	5/5/0/0	6/6/0/0
Mistral	6/3/0/3	6/3/0/3	6/3/0/3	10/5/0/5	13/6/0/7
Llama 3.3:70B	3/3/0/0	3/3/0/0	3/3/0/0	5/5/0/0	6/6/0/0
GLM-5	3/3/0/0	3/3/0/0	3/3/0/0	10/5/3/2*	6/6/0/0
Gemini 2.5-Flash	3/3/0/0	3/3/0/0	3/3/0/0	19/5/3/11**	6/6/0/0

* GLM-5 Q4: 10 extracted, 5 gold-correct, 3 extra-correct, 2 wrong.

** Gemini Q4: 19 extracted, 5 gold-correct, 3 extra-correct, 11 wrong.

Precision and Recall (Gold triples only)

Canonicalization

After using Mistral for answer and GPT for extraction, all predicates were canonicalized correctly.

Table 2: Precision, Recall, F1 for Q4 and Q5

Model	Prec (Q4)	Rec (Q4)	F1 (Q4)	Prec (Q5)	Rec (Q5)	F1 (Q5)
GPT-4.1	1.0	1.0	1.0	1.0	1.0	1.0
Mistral	0.5	1.0	0.667	0.462	1.0	0.632
Llama 3.3:70B	1.0	1.0	1.0	1.0	1.0	1.0
GLM-5	0.5	1.0	0.667	1.0	1.0	1.0
Gemini 2.5-Flash	0.263	1.0	0.417	1.0	1.0	1.0

Symbolic Verification & Logical Inconsistency

Tested on a question expecting 21 violations and 6 correct:

- GPT-4.1: 22 violations (1 extra marked as violation but explained as passed)
- Mistral: 20 violations (1 passed incorrectly)
- Llama: 21 violations
- GLM: 21 violations
- Gemini: 21 violations

Hallucination Detection

10 triples: 6 rule violations, 1 correct, 1 predicate mismatch, 1 unverifiable. All models detected all correctly.

Initial Pipeline Test

One model was tested with and without the instruction “not to answer out of context”. Errors were found; after enhancing the system and adding parts, the whole system was retested. Saved outputs are available.

Part 2: Questions

1. What are the logical inconsistencies I should detect?
2. Are there parts of the system that do not make sense?
3. Should I try another model and include it in the system?
4. What numbers do I need to include in the methodology part?
5. Should I test the system again on one model and compare performance of each module?
6. What metrics should I include when testing models on the system?

Part 3: Next Week Plan

- Finalize the system architecture.
- Finalize the models needed for each part of the system.